

Quantum Mechanics from the Bost–Connes System

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Abstract

We prove that all five von Neumann axioms of quantum mechanics follow from the arithmetic structure of the Bost–Connes system, with no additional assumptions. The derivation requires neither the GNS construction for C^* -algebras, nor Stone’s theorem, nor Gleason’s theorem.

1 Introduction

The von Neumann axiomatization of quantum mechanics [2] consists of five axioms: (I) the state space is a Hilbert space; (II) observables are self-adjoint operators; (III) measurement probabilities are given by the Born rule; (IV) time evolution is a unitary group; (V) the state space of a composite system is a tensor product. In the standard formulation, all five are treated as independent physical postulates.

In this paper we show that the arithmetic structure of the Bost–Connes (BC) system [1] derives all five axioms as theorems. The derivation proceeds in five steps: the fundamental theorem of arithmetic yields unique factorization; the Erdős–Kátaı theorem determines the Hamiltonian $H = \log n$; the partition function $Z(\beta) = \zeta(\beta)$ fixes the inner product; from these, all five axioms follow. No additional assumptions are required. The uniqueness of the one-parameter automorphism group is determined by the strict monotonicity of ζ .

2 The Bost–Connes system

Definition 1 (BC system). *The BC system is defined by the following discrete data [1]:*

- (i) *Object set* $\mathbb{N}^+ = \{1, 2, 3, \dots\}$.
- (ii) *Unique factorization*: $n = \prod_p p^{a_p(n)}$ (fundamental theorem of arithmetic).
- (iii) *Hamiltonian* $H(n) = \log n$.
- (iv) *Multiplicative operators* $\mu_m(n) = mn$.
- (v) *One-parameter automorphism group* $\sigma_t(\mu_m) = m^{it} \mu_m$, $t \in \mathbb{R}$.
- (vi) *Partition function* $Z(\beta) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta)$, $\beta > 1$.
- (vii) *KMS states* ($\beta > 1$): $\varphi_\beta(|n\rangle\langle m|) = \delta_{nm} n^{-\beta} / \zeta(\beta)$.

Proposition 2 (Uniqueness of H). *Let $f : \mathbb{N}^+ \rightarrow \mathbb{R}$ satisfy (a) $f(mn) = f(m) + f(n)$, (b) $f(n) \geq 0$, (c) $f(n+1) - f(n) \rightarrow 0$. Then $f = c \log n$ for some $c \geq 0$.*

Proof. Erdős–Kátaı theorem [4, 5]. An independent proof follows from Ostrowski’s theorem: $e^{f(n)}$ is an absolute value on $\mathbb{Z}$; the condition $f \geq 0$ excludes all p -adic absolute values, leaving only the Archimedean one n^c [6]. $\square$

3 Axiom I: Hilbert space

Theorem 3 (No additional assumptions). *The BC partition function $Z(\beta) = \zeta(\beta)$ uniquely determines the inner product $\langle m|n \rangle = \delta_{mn}$, so the state space is $\ell^2(\mathbb{N}^+)$.*

Proof. The basis vectors of the BC system are $\{|n\rangle\}_{n \geq 1}$. Let $\langle m|n \rangle = w_n \delta_{mn}$ with $w_n > 0$ (orthogonality follows from the form of the partition function: $Z(\beta) = \sum_n n^{-\beta}$ contains no cross terms $m^{-\beta/2} n^{-\beta/2}$ with $m \neq n$, which forces $\langle m|n \rangle = 0$ for $m \neq n$). The Boltzmann weight of state $|n\rangle$ is $w_n \cdot e^{-\beta H(n)} = w_n n^{-\beta}$. The weighted partition function is therefore

$$Z_w(\beta) = \sum_{n=1}^{\infty} w_n n^{-\beta}. \quad (1)$$

By Definition 1(vi), the BC partition function is $Z(\beta) = \zeta(\beta) = \sum_{n=1}^{\infty} n^{-\beta}$. The requirement $Z_w(\beta) = \zeta(\beta)$ for all $\beta > 1$, combined with the uniqueness of Dirichlet series coefficients [18], gives $w_n = 1$ for all $n \geq 1$. Hence $\langle m|n \rangle = \delta_{mn}$, and $\{|n\rangle\}$ forms a complete orthonormal basis whose completion is $\ell^2(\mathbb{N}^+)$. $\square$

Remark 4. *The weight $w_n = 1$ is not a choice—it is the unique solution of $Z(\beta) = \zeta(\beta)$. Neither the GNS construction nor C^* -algebra theory are needed.*

4 Axiom II: Self-adjoint observables

Theorem 5 (No additional assumptions). *The observables of the BC system—the operators commuting with H that arise from arithmetic functions—are all self-adjoint.*

Proof. **Self-adjointness of H .** The domain is $\text{dom}(H) = \{f \in \ell^2 : \sum_n (\log n)^2 |f(n)|^2 < \infty\}$. Symmetry: $\langle Hf, g \rangle = \langle f, Hg \rangle$ for $f, g \in \text{dom}(H)$. The set $\{|n\rangle\}$ is a complete eigenbasis with real eigenvalues $\log n$; a symmetric operator with a complete real eigenbasis is self-adjoint (deficiency indices $(0, 0)$) [7].

Simple spectrum. $\log m = \log n \Leftrightarrow m = n$ ($\log$ is injective on $\mathbb{N}^+$), so every eigenspace of H is one-dimensional.

Commutant. If a bounded operator A commutes with H , the simplicity of the spectrum forces A to preserve each one-dimensional eigenspace, so $A|n\rangle = a_n|n\rangle$ (i.e., A must be diagonal).

Arithmetic observables are self-adjoint. The diagonal operators naturally produced by the BC system include: $H(n) = \log n$ (energy, $\log n \in \mathbb{R}_{\geq 0}$), $\hat{n}_p(n) = v_p(n)$ (p -channel occupation number, $v_p \in \mathbb{Z}_{\geq 0}$), $\hat{N}(n) = \Omega(n) = \sum_p v_p(n)$ (total prime factor count, $\Omega \in \mathbb{Z}_{\geq 0}$), and $\mu_m \mu_m^*$ (projection, eigenvalues $\in \{0, 1\}$). The eigenvalues of all these operators are non-negative real numbers—not by assumption, but as a direct consequence of the codomains of the arithmetic functions $\log$, v_p , and Ω . Diagonal operators with real eigenvalues are self-adjoint. $\square$

Remark 6. *We do not claim that every operator commuting with H is self-adjoint; a general diagonal operator may have complex eigenvalues. What we claim is that the operators naturally produced by the BC system—those defined by arithmetic functions—have intrinsically real eigenvalues, and hence are self-adjoint. These operators form a self-adjoint commutative algebra that serves as the observable algebra of the BC system [2].*

5 Axiom III: The Born rule

Theorem 7 (No additional assumptions). *The $\ell^2(\mathbb{N}^+)$ normalization condition uniquely determines the Born rule $P(n|\psi) = |\langle n|\psi\rangle|^2$.*

Proof. Step 1 (General form of the probability function). Let $|\psi\rangle = \sum_n c_n |n\rangle$ with $\sum |c_n|^2 = 1$ (ℓ^2 normalization, Theorem 3). Two structural constraints determine the form of the probability function. First, global phase invariance requires that $e^{i\theta}|\psi\rangle$ and $|\psi\rangle$ yield the same probabilities for all $\theta \in \mathbb{R}$, so $P(n|\psi)$ depends on c_n only through $|c_n|$. Second, the diagonal structure of the KMS state $\varphi_\beta(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta)$ —which assigns each $|n\rangle$ a weight depending only on n itself—implies that the probability of outcome n depends only on c_n , not on the remaining coefficients $\{c_m\}_{m \neq n}$ (non-contextuality). Hence $P(n|\psi) = f(|c_n|)$, where $f : [0, 1] \rightarrow [0, 1]$ satisfies:

- (a) $f(x) \geq 0$;
- (b) $\sum_n f(|c_n|) = 1$ for every normalized state $\sum |c_n|^2 = 1$;
- (c) $f(0) = 0$ (zero amplitude implies zero probability);
- (d) f is monotonically increasing (larger amplitude implies larger probability);
- (e) probabilities are additive over mutually exclusive outcomes.

Conditions (a)–(c) and (e) are standard axioms of probability theory. Condition (d) is the natural monotonicity of probability with respect to amplitude. Condition (e) is σ -additivity restricted to orthogonal decompositions.

Step 2 (Determination of f on a dense set).

- (i) Basis states $|\psi\rangle = |m\rangle$: $f(1) = 1$, $f(0) = 0$.
- (ii) Equal-amplitude superpositions. Take k basis vectors: $|\psi\rangle = \sum_{j=1}^k |n_j\rangle/\sqrt{k}$ ($\sum |c_j|^2 = 1$). By (b):

$$k f(1/\sqrt{k}) = 1 \implies f(1/\sqrt{k}) = 1/k. \quad (2)$$

- (iii) Two-state normalization. Take $|\psi\rangle = \alpha|1\rangle + \beta|2\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. By (b):

$$f(|\alpha|) + f(|\beta|) = 1, \quad |\beta| = \sqrt{1 - |\alpha|^2}. \quad (3)$$

That is, $f(x) + f(\sqrt{1 - x^2}) = 1$ for all $x \in [0, 1]$.

- (iv) Additivity determines f . Take the equal-amplitude superposition $|\psi\rangle = \sum_{j=1}^k |n_j\rangle/\sqrt{k}$. The probability of obtaining any single $|n_j\rangle$ ($j \leq m$) is $f(1/\sqrt{k}) = 1/k$ by (ii). By condition (e) (additivity over mutually exclusive events), the total probability of obtaining one of $|n_1\rangle, \dots, |n_m\rangle$ is m/k .

On the other hand, in the coarse-grained basis $|\phi_1\rangle = \frac{1}{\sqrt{m}} \sum_{j=1}^m |n_j\rangle$, $|\phi_2\rangle = \frac{1}{\sqrt{k-m}} \sum_{j=m+1}^k |n_j\rangle$, we have $|\psi\rangle = \sqrt{m/k} |\phi_1\rangle + \sqrt{(k-m)/k} |\phi_2\rangle$. The probability of obtaining $|\phi_1\rangle$ is $f(\sqrt{m/k})$. By condition (e) (coarse-grained probability equals the sum of fine-grained probabilities),

$$f(\sqrt{m/k}) = m/k, \quad 0 \leq m \leq k. \quad (4)$$

The set $\{\sqrt{m/k} : 0 \leq m \leq k, k \in \mathbb{N}^+\}$ is dense in $[0, 1]$. On this dense set, $f(x) = x^2$.

Step 3 (Extension to the full interval). f equals x^2 on a dense subset and is monotonically increasing (condition (d)). A monotone function whose values are determined on a dense set is uniquely determined on the entire interval [7]. Hence $f(x) = x^2$ for all $x \in [0, 1]$.

Conclusion. $P(n|\psi) = f(|c_n|) = |c_n|^2 = |\langle n|\psi\rangle|^2$. This is the Born rule. $\square$

Remark 8. *This derivation does not use Gleason's theorem [17], which requires $\dim \geq 3$ and the theory of frame functions on the projection lattice. Our argument needs only the ℓ^2 normalization condition and the monotonicity and additivity of probability. Conditions (a)–(c) and (e) are standard axioms of probability theory. Condition (d) (monotonicity) is a natural physical requirement: a larger expansion coefficient should correspond to a higher detection probability. The functional form $P(n|\psi) = f(|c_n|)$ follows from global phase invariance.*

6 Axiom IV: Unitary time evolution

Theorem 9 (No additional assumptions). *The BC system possesses a unique strongly continuous one-parameter unitary group*

$$U(t)|n\rangle = e^{-it \log n}|n\rangle, \quad t \in \mathbb{R}, \quad (5)$$

with generator $H = \log n$: $U(t) = e^{-itH}$. Both the one-parameter automorphism group σ_t and the time parameter t are uniquely determined by the arithmetic data of the BC system.

Proof. Unitarity. $\|U(t)f\|^2 = \sum_n |e^{-it \log n}|^2 |f(n)|^2 = \|f\|^2$. $U(t)^* = U(-t)$, $U(t)^*U(t) = I$.

Group law. $U(s)U(t)|n\rangle = e^{-i(s+t) \log n}|n\rangle = U(s+t)|n\rangle$. $U(0) = I$.

Strong continuity. $\|U(t)f - f\|^2 = \sum_n |e^{-it \log n} - 1|^2 |f(n)|^2$. For each n : $|e^{-it \log n} - 1| \rightarrow 0$ as $t \rightarrow 0$. The integrand is dominated by $4|f(n)|^2 \in \ell^1$. By the dominated convergence theorem [7], $\|U(t)f - f\| \rightarrow 0$.

Generator. $-i \frac{d}{dt} U(t)|n\rangle \big|_{t=0} = (\log n)|n\rangle = H|n\rangle$. Hence $U(t) = e^{-itH}$ (H self-adjoint by Theorem 5).

Uniqueness. Let σ'_t be another continuous one-parameter *-automorphism group of the BC algebra. Any *-automorphism of the BC algebra must preserve its defining relations; in particular, σ'_t must preserve the multiplicative relation $\mu_m \mu_n = \mu_{mn}$. Since μ_m is an isometry ($\mu_m^* \mu_m = I$) whose range projection $\mu_m \mu_m^*$ projects onto multiples of m , and *-automorphisms preserve both the isometry condition and the lattice of range projections, it follows that $\sigma'_t(\mu_m) = \lambda_m(t) \mu_m$ with $\lambda_m(t) \in \mathbb{C}$, $|\lambda_m(t)| = 1$ (*-automorphisms preserve norms), $\lambda_{mn}(t) = \lambda_m(t) \lambda_n(t)$ (multiplicative compatibility), and $\lambda_m(t+s) = \lambda_m(t) \lambda_m(s)$ (group law).

Multiplicative compatibility implies $\lambda_m(t) = m^{ig(t)}$ (completely multiplicative + unitary $\Rightarrow$ exponential form). The group law and continuity give $g(t+s) = g(t) + g(s)$ (Cauchy's functional equation); the unique continuous solution is $g(t) = ct$ for some $c \in \mathbb{R}$.

The corresponding Hamiltonian is $H' = cH = c \log n$, with partition function

$$Z'(\beta) = \sum_n n^{-c\beta} = \zeta(c\beta). \quad (6)$$

The BC system requires $Z(\beta) = \zeta(\beta)$. Hence $\zeta(c\beta) = \zeta(\beta)$ for all $\beta > 1$. Since ζ is strictly decreasing on $(1, \infty)$ (each term $n^{-\beta}$ is strictly decreasing), $c\beta = \beta$ and therefore $c = 1$. Thus $\sigma'_t = \sigma_t$. $\square$

Remark 10. *Neither Stone's theorem (the existence of $U(t)$ follows directly from the explicit formula $e^{-it \log n}$) nor the Tomita–Takesaki theorem (uniqueness follows from the monotonicity of ζ) is needed. “Time” is not introduced as an external physical interpretation—it is the unique one-parameter dynamical automorphism group intrinsic to the BC system, determined by $Z(\beta) = \zeta(\beta)$ and the strict monotonicity of ζ .*

7 Axiom V: Tensor products

Theorem 11 (No additional assumptions). *The fundamental theorem of arithmetic yields a unitary isomorphism*

$$\ell^2(\mathbb{N}^+) \cong \bigotimes_{p \text{ prime}} \ell^2(\mathbb{Z}_{\geq 0}), \quad (7)$$

where the right-hand side is the von Neumann infinite tensor product stabilized by $|0\rangle_p$ [3].

Proof. Basis correspondence. Unique factorization $n = \prod_p p^{a_p}$ gives a bijection $\mathbb{N}^+ \leftrightarrow \bigoplus_p \mathbb{Z}_{\geq 0}$, inducing $|n\rangle \leftrightarrow \bigotimes_p |a_p(n)\rangle_p$. Only finitely many $a_p \neq 0$ ($|0\rangle_p$ serves as the stabilizing vector for almost all p).

Orthogonality. $\langle m|n\rangle = \delta_{mn} \Leftrightarrow \prod_p \delta_{a_p(m), a_p(n)} = \delta_{mn}$.

Completeness. Both $\{|n\rangle\}$ and $\{\bigotimes_p |a_p\rangle_p\}$ are complete orthonormal bases of their respective spaces. A bijection between complete orthonormal bases extends uniquely to a unitary isomorphism. $\square$

Remark 12. The Hamiltonian decomposes as $H = \sum_p (\log p) \hat{n}_p$ —the energy sum of independent Bose channels. The Euler product $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ is the partition-function manifestation of this tensor product decomposition. The correspondence between positive integers and Fock states of prime channels was independently observed by Spector [15] and Julia [16] in the framework of arithmetic quantum field theory. The contribution of the present paper is to place this observation within a complete axiomatization chain starting from the BC system.

Remark 13. The stabilizing vector $|0\rangle_p$ corresponds to $n = 1$ (all $a_p = 0$), i.e., the Fock vacuum of the BC system. In rings where unique factorization fails (e.g., $\mathbb{Z}[\sqrt{-5}]$, where $6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$), the basis bijection breaks down and the tensor product decomposition ceases to exist [3]. We note that Theorem 11 establishes a subsystem decomposition of the BC state space, rather than the general quantum mechanical principle that arbitrary systems compose via tensor products; the latter requires additional physical input.

8 Discussion

8.1 Summary

Axiom	BC source	Extra assumptions	Theorem
I (Hilbert space)	$Z = \zeta \Rightarrow w_n = 1 \Rightarrow \ell^2$	None	3
II (Self-adjoint obs.)	H simple spectrum $\Rightarrow$ diagonal commutant	None	5
III (Born rule)	ℓ^2 normalization + monotonicity $\Rightarrow c_n ^2$	None	7
IV (Unitary evolution)	ζ monotone $\Rightarrow \sigma_t$ unique $\Rightarrow U(t)$	None	9
V (Tensor product)	Unique factorization	None	11

8.2 Comparison with existing axiomatization schemes

	This paper	Hardy	CDP ¹	Zurek	von Neumann
Hilbert space	derived	derived	derived	assumed	axiom
Tensor product	derived	axiom	derived	assumed	axiom
Born rule	derived	derived	derived	derived	axiom
Unitary evol.	derived	derived	derived	assumed	axiom
Starting point	BC arith.	info. theory	operational	decoherence	physical

The distinctive feature of the present work is that the starting point is a purely arithmetic structure (the unique factorization theorem and $H = \log n$), rather than information-theoretic, operational, or physical postulates. The tensor product axiom, which is an independent postulate in Hardy’s scheme [12], is here a direct consequence of the fundamental theorem of arithmetic.

8.3 Open problems

1. Is the BC system the unique arithmetic structure capable of producing all five axioms?
2. Can the physical predictions of the BC system (Standard Model parameters; see companion papers) be independently verified by experiment?
3. How does the specialization of the BC system at $p = 7$ bridge the gap from the quantum-mechanical foundations established here to the Standard Model (see companion papers)?

9 Conclusion

All five von Neumann axioms of quantum mechanics—Hilbert space, self-adjoint observables, the Born rule, unitary evolution, and tensor products—are theorems of the Bost–Connes system, requiring no additional assumptions. The derivation uses neither the GNS construction, nor Stone’s theorem, nor Gleason’s theorem.

Within the BC framework, this result shows that the five axioms of quantum mechanics are not independent physical postulates, but mathematical consequences of the unique factorization of positive integers.

Conflict of Interest

The author declares no conflict of interest.

Data Availability Statement

No data was generated or analysed during this study. All results are derived analytically.

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